VolatilityForge: Market Volatility Forecasting Using Econometric and Machine Learning Models

VolatilityForge Project Research Report

SPY and CBOE VIX | Expanding-window walk-forward evaluation | QLIKE and MSE

Research status: Development-period evaluation completed; prospective validation remains outstanding.

Abstract

Forecasting financial-market volatility is an important problem in quantitative finance, risk management, and portfolio construction. Volatility is time-varying, tends to cluster during turbulent market periods, and cannot be observed directly in the same way as asset prices. This project, VolatilityForge, develops a reproducible research pipeline for forecasting realized variance using statistical, econometric, and machine learning approaches.

The project uses daily SPY market data and the CBOE Volatility Index (VIX) as an implied-volatility reference. It generates next-session and five-session-ahead realized-variance targets, constructs historical volatility features, and evaluates forecasts using expanding-window walk-forward procedures. The implemented forecasting approaches include EWMA, GARCH, GJR-GARCH, HAR-RV, LightGBM, and a VIX-implied benchmark.

Forecast performance is assessed using QLIKE and MSE, alongside Diebold-Mariano comparisons, calibration reports, volatility-regime analysis, and Model Confidence Set (MCS) procedures. A separate portfolio experiment evaluates volatility-targeting strategies using transaction costs, risk-adjusted performance measures, drawdown, turnover, and a buy-and-hold benchmark.

In the current development-period MCS analysis, LightGBM is the sole retained model for both one-day and five-day forecasts under the reported settings. The strategy results illustrate the trade-off between return and risk reduction: volatility targeting lowers realized portfolio volatility and drawdown relative to buy-and-hold in the displayed comparison, while also producing lower returns. These findings are historical development results rather than evidence of future performance. A prospective final evaluation has not yet been conducted.

Index Terms - realized variance, volatility forecasting, SPY, VIX, QLIKE, LightGBM, walk-forward evaluation, volatility targeting.

Research disclaimer: VolatilityForge is an educational and research project. Historical forecast and backtest results do not guarantee future performance and are not financial advice.

1. Introduction

1.1 Background

Financial-market volatility changes over time. Quiet market periods can be followed by sharp increases in uncertainty, and large market movements often occur in clusters. A forecasting model that estimates future variance can support portfolio-risk management and position sizing.

Volatility forecasting differs from price forecasting because the target is uncertainty about future returns rather than the direction of those returns. Forecast quality must therefore be assessed using loss functions and statistical procedures appropriate for variance forecasts.

Traditional econometric models provide interpretable volatility dynamics, while machine learning models can capture nonlinear relationships between historical features and future realized variance. Implied volatility offers another perspective by reflecting option-market expectations, although it is not identical to subsequently realized variance.

1.2 Research objectives

The project has the following objectives:

1. Build a reproducible SPY and VIX data-processing pipeline.
2. Construct leakage-aware realized-variance targets at one-day and five-day horizons.
3. Implement statistical and machine learning forecasting approaches.
4. Compare forecasts using consistent samples and volatility-forecasting metrics.
5. Examine forecast differences using statistical comparisons, calibration, and regime analysis.
6. Evaluate volatility-targeting strategies with transaction costs.
7. Provide documented experiment outputs and an interactive research dashboard.

1.3 Research questions

The project investigates three central questions:

- How do the implemented statistical, econometric, and machine learning models compare when forecasting realized variance?
- Does the relative performance of the models vary across forecast horizons and market-volatility regimes?
- How does forecast-driven volatility targeting change portfolio returns, volatility, and drawdown relative to a buy-and-hold benchmark?

These questions are evaluated using historical development data. They are not claims about performance on unseen future observations.

2. Data and Target Construction

2.1 Market data

The project uses daily SPY observations as the primary market series and VIX as an implied-volatility reference. The data pipeline downloads the configured market data, validates the required OHLCV fields, saves raw data in Parquet format, aligns SPY and VIX observations by date, and generates processed volatility features.

The principal pipeline outputs are:

- `data/raw/spy.parquet`
- `data/raw/vix.parquet`
- `data/processed/spy_vix.parquet`
- `data/processed/volatility_features.parquet`

The pipeline uses the configured project settings in `configs/config.yaml`. Rebuilding the data requires network access to the market-data provider.

Historical observations can be revised by the provider. Consequently, rerunning the pipeline does not guarantee bit-for-bit reproduction of previously downloaded data.

2.2 Realized-variance targets

The forecasting framework supports two horizons.

One-day target: next-session squared return, representing a daily realized-variance proxy.

Five-day target: cumulative squared returns over the next five trading sessions, representing cumulative realized variance across that horizon.

The project models these targets using a leakage-aware workflow. A forecast generated after the close on date t must not use future returns or target observations that have not matured by that forecast origin.

For the five-day horizon, training labels must be fully observed before they can be used for fitting or residual estimation. The documented protocol applies a strict target-maturity rule to prevent overlap between unavailable future information and model training.

$$RV_{t+1} = r_{t+1}^{2} \tag{1}$$

$$RV_{t+1:t+5} = sum_{j=1}^{5} r_{t+j}^{2} \tag{2}$$

The equations formalize the next-session squared-return proxy and the cumulative five-session realized-variance target described in the project record.

2.3 Feature engineering

The implemented feature pipeline includes historical volatility information and range-based estimators derived from OHLC data. The existing feature set includes:

- Historical realized variance over short and longer windows.
- Overnight price gaps.
- Parkinson variance estimator.
- Garman-Klass variance estimator.
- Rogers-Satchell variance estimator.
- VIX close and related implied-volatility information.

HAR-RV uses daily, weekly, and monthly realized-variance components with windows of 1, 5, and 22 trading days. The objective is to represent volatility persistence at different time scales.

Features are intended to use information available by the forecast origin. The evaluation protocol requires that future target information remain excluded from feature construction and model fitting.

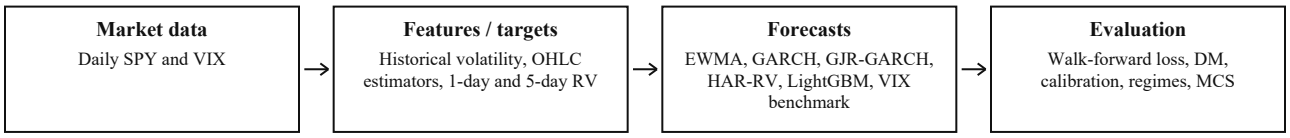


Figure 1. Schematic of the data, target, forecasting, and evaluation components described in the supplied project record.

3. Forecasting Methodology

3.1 EWMA

Exponentially Weighted Moving Average (EWMA) estimates current variance by assigning exponentially decreasing weights to older squared returns. Recent observations therefore contribute more strongly to the estimate than distant observations.

EWMA serves as a simple, responsive benchmark against which more complex models can be compared.

3.2 GARCH

The Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model represents conditional variance using previous squared innovations and previous conditional variance. It captures volatility clustering and persistence.

GARCH provides an econometric benchmark for assessing whether more flexible approaches improve forecast accuracy.

3.3 GJR-GARCH

GJR-GARCH extends the conditional-variance specification to allow asymmetric responses to positive and negative return shocks. This is relevant when negative market movements are associated with different volatility responses than positive movements.

Its inclusion allows the evaluation to compare a symmetric conditional-variance specification with an asymmetric alternative.

3.4 HAR-RV

The Heterogeneous Autoregressive model of Realized Volatility (HAR-RV) represents volatility dynamics using realized-variance components at daily, weekly, and monthly scales.

The model is a structured benchmark for volatility persistence and multiscale behavior. Both one-day and five-day forecasting runners are included in the project.

A separate experimental branch evaluates Duan's smearing correction for the five-day HAR-RV model. That correction is not part of the production HAR-RV forecasting implementation.

3.5 LightGBM

LightGBM is the project's implemented gradient-boosting machine learning model. It can capture nonlinear relationships and interactions between engineered volatility features and future realized-variance targets.

Separate runners support one-day and five-day forecasting. The evaluation uses the project's chronological forecasting protocol rather than random train-test splitting.

3.6 VIX-implied benchmark

The VIX-implied benchmark converts the VIX level into an annualized variance proxy and scales it to daily and five-day horizons.

This is a reference benchmark rather than a direct observation of future realized variance. Implied volatility reflects option-market pricing and expectations, while realized variance is calculated from subsequent returns. The distinction is important when interpreting comparisons.

3.7 Models outside the implemented scope

Ridge, Lasso, XGBoost, and a HAR-RV residual hybrid are not implemented in the current model package. They are not included as completed models in this report. LSTM and multi-asset generalization are also not claimed as completed deliverables.

4. Evaluation Design

4.1 Walk-forward evaluation

The project uses expanding-window walk-forward evaluation. Models are fitted using historical data available at each forecast origin, and forecasts are evaluated against subsequent realized outcomes.

The evaluation configuration specifies:

- Evaluation method: expanding window.
- Development evaluation start: January 2022.
- Primary metric: QLIKE.
- Secondary metric: MSE.
- Minimum training observations: 252.
- Re-estimation interval: 21 observations.

The development period has been used for model comparisons, experimental corrections, regime analysis, and strategy assessment. It must therefore not be described as an untouched final holdout.

4.2 QLIKE and MSE

QLIKE is the primary forecast-comparison metric because it is suitable for comparing variance forecasts and penalizes forecast errors in a way that differs from squared-error loss. Lower values indicate better performance under the project's reported convention.

MSE measures average squared differences between forecast and realized target values. It provides a complementary view of forecast error but can respond strongly to large errors.

The project generates model-comparison reports for both one-day and five-day horizons. Comparisons should use common evaluation samples so that differences are not driven by inconsistent dates or observation counts.

4.3 Diebold-Mariano comparisons

The project includes Diebold-Mariano (DM) forecast-comparison reports with sensitivity analysis over Newey-West heteroskedasticity-and-autocorrelation-consistent lag choices. The documented protocol specifies a Harvey-Leybourne-Newbold correction and multiple-comparison adjustment.

These procedures provide evidence about relative forecast losses under specified assumptions. Their conclusions are conditional on the selected sample, loss function, dependence corrections, and comparison design.

4.4 Calibration and volatility regimes

Calibration reports are generated separately for one-day and five-day forecasts. Regime reports compare model performance across different realized-volatility conditions.

These analyses complement aggregate forecast metrics. A model with favorable average loss may still perform inconsistently in particular regimes or on individual dates.

4.5 Model Confidence Set

The project uses `arch.bootstrap.MCS` for the reported Model Confidence Set analysis. The saved sensitivity report evaluates block sizes of 5, 10, and 22 under the documented procedure.

MCS results identify a set of models that are not eliminated by the procedure at the selected significance level. They should not be interpreted as proof that a retained model is universally superior or will remain best on future data.

5. Forecast Evaluation Results

5.1 Model Confidence Set results

The generated MCS reports contain 1,200 one-day observations and 1,198 five-day observations in the reported development evaluation.

Table 1: Model Confidence Set result by forecast horizon.

Forecast horizon	Evaluation observations	Retained model
One day	1,200	LightGBM
Five days	1,198	LightGBM

Under the reported MCS settings, LightGBM was the only retained model at all three tested block sizes for each horizon.

The reported mean QLIKE values were:

Table 2: Mean QLIKE by model and forecast horizon.

Model	One-day mean QLIKE	Five-day mean QLIKE
LightGBM	1.439196	0.296152
VIX-implied	1.520443	0.378811
EWMA	1.532757	0.394436
GARCH	1.664255	0.487107
GJR-GARCH	1.665692	0.484708
HAR-RV	1.527426	0.457113

The table reports development-period results only. The QLIKE values for the two forecast horizons should be interpreted within their respective target definitions and evaluation setups; they are not directly comparable as if they measured the same prediction task.

5.2 Interpretation

LightGBM has the lowest reported mean QLIKE among the listed models at both horizons, and the MCS procedure retains it as the sole model under the tested settings.

This is evidence in favor of LightGBM within the evaluated development sample and the specified comparison procedure. It does not establish that LightGBM will outperform the other models on future data.

The VIX-implied benchmark provides a useful reference but does not replace a model trained directly against realized-variance targets. The statistical models remain important baselines because they provide interpretable and established volatility dynamics.

5.3 Experimental HAR-RV smearing correction

A separate experiment evaluates a smearing correction for five-day HAR-RV forecasts using historical out-of-sample log residuals whose five-day targets have fully matured before each forecast origin.

The experiment report records the following results:

Table 3: Five-day HAR-RV smearing experiment.

Metric	Baseline HAR-RV	OOS-smearing HAR-RV
Mean QLIKE	0.457635	0.372767
MSE	1.017897 x 10^-6	9.917346 x 10^-7
Observations	1,196	1,196

The experiment's documented evaluation period runs from January 3, 2022, through October 8, 2026.

Although aggregate QLIKE and MSE improved in this experiment, the gains were not uniform. The report records deterioration in the low-realized-variance regime, and QLIKE improved on 496 of 1,196 individual dates.

For these reasons, the smearing correction remains experimental and has not been adopted in the production HAR-RV implementation. The experiment does not establish statistical significance or future performance.

6. Volatility-Targeting Strategy Evaluation

6.1 Strategy design

The strategy module converts volatility forecasts into portfolio exposure. The configured baseline uses a 10% annualized volatility target, maximum leverage of 1.0, and transaction costs of 5 basis points per unit of turnover.

The backtest applies a forecast generated after the close on date t to the following session's return. Exposure is long-only and capped by the configured maximum leverage. Transaction costs are charged on absolute changes in exposure.

The reports include gross and net strategy returns, annualized return and volatility, zero-risk-free-rate Sharpe ratio, maximum drawdown, average exposure, turnover, and a same-period gross buy-and-hold benchmark.

Sensitivity analysis evaluates five forecasting models at target-volatility levels of 8%, 10%, and 12%, with transaction-cost assumptions of 0, 5, 10, and 20 basis points.

6.2 Illustrative strategy comparison

The following results come from the reported development-period strategy output.

Table 4: Illustrative development-period volatility-targeting comparison.

Metric	EWMA strategy, net	Buy-and-hold, gross
Observations	1,202	1,202
Total return	38.89%	70.95%
Annualized return	7.13%	11.90%
Annualized volatility	10.25%	17.27%
Sharpe ratio, zero risk-free rate	0.723	0.737
Maximum drawdown	-13.19%	-24.50%
Average exposure	0.705	1.000
Average turnover	0.0199	0.0000

The EWMA strategy's net-return series deducts transaction costs. The buy-and-hold benchmark is gross and excludes transaction costs, so the comparison is not cost-matched.

6.3 Interpretation

In this sample, the EWMA volatility-targeting strategy has lower annualized volatility and a smaller maximum drawdown than buy-and-hold. Its annualized return and reported Sharpe ratio are slightly lower.

The results illustrate the economic trade-off between reducing portfolio exposure during lower-volatility allocations and participating fully in market returns. The strategy is not demonstrated to dominate buy-and-hold across all performance dimensions.

These historical results do not establish live executability, profitability after all real-world trading frictions, or future risk reduction. Transaction-cost assumptions, execution, exposure timing, and the development-period nature of the evaluation all limit the conclusions.

7. Reproducibility and Implementation

7.1 Repository organization

The repository separates configuration, data, source code, tests, reports, documentation, notebooks, scripts, and dashboard components.

Key components include:

- `configs/config.yaml` - project and evaluation settings.
- `src/data_loader.py` and the data-pipeline runner - market-data ingestion.
- `src/feature_engineering.py` - features and forward targets.
- `src/models/` - forecasting model implementations.
- `src/compare_forecasts.py` and `src/compare_five_day_forecasts.py` - forecast comparisons.
- MCS and strategy runners - statistical and economic evaluation outputs.
- `reports/` - generated result tables.
- `docs/evaluation_protocol.md` - evaluation rules and leakage controls.
- `DECISIONS.md` - recorded project decisions.
- `dashboard/app.py` - Streamlit research dashboard.

7.2 Rebuilding the data pipeline

With the project's virtual environment activated, run from the repository root:

```
python -m src.run_data_pipeline
```

This command downloads the configured SPY and VIX data, validates and saves raw data, aligns dates, and rebuilds the processed features and targets. Network access is required.

7.3 Strategy report reproduction

After the required processed features and forecast files are present, the strategy outputs can be regenerated using:

```
python -m src.run_strategy_backtest
python -m src.run_strategy_report
python -m src.run_strategy_sensitivity
```

These commands reuse saved forecasts and processed data. They do not independently rebuild the entire pipeline or retrain all forecasting models.

7.4 Dashboard

The dashboard launch command documented by the project is:

```
python -m streamlit run dashboard/app.py
```

The dashboard provides overview, model-comparison, volatility-regime, and strategy-backtest views. The existence of the dashboard entry point is documented; a successful interactive launch should be confirmed separately before claiming that every dashboard view has been manually verified.

7.5 Verification status

The final local verification run reported:

- Ruff lint: all checks passed.
- Automated tests: 314 passed.
- Git whitespace check: passed, with line-ending warnings only.
- Git commit: `c028fc5`, with message `Finalize VolatilityForge research pipeline`.
- Git working tree: clean after the commit.

These results confirm that the existing automated suite and lint checks passed for the committed code. They do not establish correctness for every possible input, market regime, or future observation.

8. Limitations and Future Work

8.1 Development data versus final evaluation

The period beginning in 2022 has already been used for model comparison, experimental corrections, regime analysis, and strategy backtests. It is therefore development data, not an untouched final holdout.

A prospective evaluation period should only be defined after the model specifications, feature definitions, hyperparameters, and evaluation procedure have been frozen. That future period should not be used for further model selection or tuning if it is to serve as an independent final evaluation.

8.2 Model coverage

The implemented model set includes EWMA, GARCH, GJR-GARCH, HAR-RV, LightGBM, and the VIX-implied benchmark. Ridge, Lasso, XGBoost, and a HAR-RV residual hybrid remain unimplemented. The current results should not be generalized to those models.

8.3 Data and market limitations

Historical data providers may revise observations. The current project focuses on SPY and VIX and does not establish generalization to other assets or markets. The reported strategy evaluation uses specified transaction-cost assumptions rather than a complete simulation of real trading, liquidity, market impact, and execution constraints.

8.4 Statistical interpretation

The MCS results depend on the chosen loss function, evaluation sample, bootstrap settings, and assumptions. Regime-level and experimental findings can differ from aggregate metrics. The smearing experiment's mixed regime behavior demonstrates why improvements in an aggregate metric should not automatically be treated as universally beneficial.

8.5 Future work

The next research steps should prioritize independent validation and reproducibility rather than indiscriminate feature expansion:

1 Freeze the existing evaluation protocol and model specifications.
2 Define and preserve an untouched prospective evaluation period.
3 Evaluate frozen models on that period without tuning against its results.
4 Verify the dashboard and documented reproduction commands in a clean environment.
5 Consider additional models or multi-asset evaluation only if justified by a clearly defined research question.

9. Conclusion

VolatilityForge provides a reproducible foundation for studying daily realized-variance forecasting and forecast-driven volatility targeting. It combines an SPY/VIX data pipeline, leakage-aware one-day and five-day targets, statistical and machine learning forecasts, forecast-comparison procedures, regime analysis, and transaction-cost-aware strategy reports.

Within the current development evaluation, LightGBM has the lowest reported mean QLIKE for both forecast horizons and is the only model retained by the reported MCS analysis. The strategy comparison shows that volatility targeting can reduce realized portfolio volatility and maximum drawdown in the evaluated sample, but this comes with lower returns than the gross buy-and-hold benchmark in the displayed EWMA comparison.

These conclusions remain conditional on the historical development sample and the specified evaluation methods. The project has not yet completed a prospective untouched final evaluation, and several candidate models remain outside the implementation.

The appropriate conclusion is that VolatilityForge is a tested and committed research implementation with documented empirical results-not a guarantee of future forecasting accuracy or trading profitability.

References and project artifacts

This report is based on the project's existing implementation, configuration, evaluation protocol, decision log, and generated CSV reports. The main supporting artifacts are:

- `docs/evaluation_protocol.md`
- `DECISIONS.md`
- `reports/volatility_model_comparison.csv`
- `reports/five_day_model_comparison.csv`
- `reports/dm_test_sensitivity.csv`
- `reports/five_day_dm_sensitivity.csv`
- `reports/one_day_calibration.csv`
- `reports/five_day_calibration.csv`
- `reports/mcs_one_day.csv`
- `reports/mcs_five_day.csv`
- `reports/mcs_sensitivity.csv`
- `reports/regime_comparison_1d.csv`
- `reports/regime_comparison_5d.csv`
- `reports/strategy_performance.csv`
- `reports/strategy_sensitivity.csv`

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